

Physics-Reward Finetuning of a Diffusion Model for Sparse Flow Reconstruction

DDPO on 2D Kolmogorov turbulence — spectral gains, the placement ceiling, OOD extrapolation, and the resolution at which the reward pays off

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Abstract. We finetune a native 256×256 denoising diffusion model with reference-free physics rewards (DDPO/PPO) to reconstruct 2D Kolmogorov vorticity fields from sparse observations. The reward reliably improves every *spectral* statistic it optimizes — high-k energy retention, per-scale spectrum, effective resolution k^* , and PDE residual — on held-out data and without reward hacking. It does *not* improve pointwise reconstruction at the operating density (1024 random points, 1.56%): the model adds correctly-shaped energy in the wrong places because fine-scale phase is underdetermined by the sparse input. Three results follow. (i) The finetuning transfers out-of-distribution to $Re=2000$ using anchors *extrapolated* from $Re=500$ and $Re=1000$, requiring no high-resolution target data (+0.075 retention, matching/beating the measured-anchor run). (ii) The placement ceiling is provably information-limited: energy-placement correlation rises $0.18 \rightarrow 0.34 \rightarrow 0.81$ as observed points increase $256 \rightarrow 1024 \rightarrow 4096$. (iii) DDPO begins to pay off at $\sim 3,000$ observed points ($\sim 4.5\times$ downsampling), where placement crosses ~ 0.6 while a genuine high-k amount deficit remains for the reward to fill without a placement penalty. Multi-phase sampling ($K=3$) is an orthogonal, inference-time lever that halves the residual and gives the first real pointwise gain.

1 Problem setup

Model. Native 256×256 unconditional DDPM (ch_mult [1,1,1,2], $T=1000$), trained on 256×256 vorticity-triplet fields. It is *not* a $64 \rightarrow 256$ upsampler; the task is **sparse-observation reconstruction**: keep a small set of pixels, nearest-neighbour-fill to a full 256×256 field, and refine via SDEdit (noise to $t=100$, denoise). ‘Downsampling factor’ sets observation sparsity, not tensor size.

Real-task degradation. 1024 random collocation points = 1.56% ($1/64$) of pixels, per-sequence random masks, NN-filled (Voronoi). Aggressive. **DDPO:** 4-chip pmap PPO, per-input advantage baseline (mandatory — cancels the regime-anchor offset), KL-to-base, sampling temperature 1.5, batch 32 (8/chip HBM ceiling for backprop through the denoising chain).

Reward (reference-free, anchored to regime statistics): dominant spectral terms *spec_highk* ($k \geq 32$, weight 3.0) and *spec*; a total-entrophy *energy* guard; a hinged PDE-residual term; plus two structural terms studied here (*pde_local*, *align*).

2 In-distribution $Re=1000$: spectral win, pointwise flat

Best models on the full val(32–35)+test(36–39) set, change vs the frozen base (base retention 0.64–0.69). Every spectral metric improves; pointwise MSE is flat.

model	retention Δ	k^* Δ	residual Δ	MSE Δ
pde_local (A)	+0.071	+8	−9.5	~flat
plain DDPO (B)	+0.096	+14	−9.0	~flat
align (C, best)	+0.098	+20 (44→64)	−8.8	flat (<0.5%)

Residual/error maps are visually unchanged and $\text{corr}(\text{recon} - \text{base}, \text{GT} - \text{base}) \approx 0.02$: what is added is uncorrelated with what is truly missing.

3 OOD $Re=2000$ with extrapolated anchors — headline

Finetuning the $Re=1000$ model toward $Re=2000$. The extrapolated anchor is built from $\{Re=500, 1000\}$ only (spectrum via the dissipation-scale law $k_d \propto Re^{1/2}$, saturated entrophy, residual floor via the hi-k entrophy ratio); it is within $d_{\text{spec}} \approx 0.038$ of the measured anchor. Full 20-sequence eval (val 0–9 / test 10–19).

anchor	retention Δ	residual Δ	MSE	target-regime data
measured Re=2000	+0.066 / +0.067	-7.2	flat	yes (real stats)
extrapolated {500,1000}	+0.073 / +0.074	-3.6	flat	NONE

The extrapolated-anchor run tracked and slightly beat the measured one across all 200 iterations. **DDPO can finetune toward an unseen regime with no high-resolution target data.** Capacity caveat: the base is missing ~73% of Re=2000 hi-k energy; DDPO closes ~1/12 of that deficit.

4 Can any reward fix placement?

Placement metric: $\text{corr}(E_{\text{hik}}(\text{recon}), E_{\text{hik}}(\text{GT}))$, frozen at ≈ 0.44 for the base. Three controlled experiments, same seed and configuration.

experiment	outcome	placement move
pde_local (worst-region residual)	negative: no extra cleanup, costs retention	-0.004
strain-placement (energy→strain regions)	killed at diagnostic (headroom -0.18)	—
strain-orientation → align	best structural term; best k^*, no cost	+0.005

The *align* reward pushes the small-scale gradient to lock onto the compressive strain axis (GT statistic 0.289; isotropic 0.5; recon 0.394). It is the only reward to move placement positively, but only ~7% of the gap — policy-gradient exploration is isotropic, so well-oriented samples are rare. **Conclusion: at 1024 points, placement will not move materially by reward.**

5 Input baseline and multi-phase sampling

Does the reconstruction beat the NN-filled input? Enormously. The Voronoi input is physically catastrophic; the model removes essentially all of it. **Does multi-phase ($K=3$, $S=[150,100,50]$) help?** Yes — the first real pointwise gain (val split shown).

field	hik_ret	PDE residual	MSE
NN-fill input	7.45× (spurious)	10,287 (~800× floor)	0.242
align, single chain	0.77	51	0.156
align, K=3 multi-phase	0.76	29	0.145

Versus the single chain, $K=3$ halves the residual ($51 \rightarrow 29$), lowers MSE ~7%, holds retention, and improves spatial placement $0.35 \rightarrow 0.41$ (~10× the align reward). Iterative renoise/denoise re-projects onto the physical manifold — a sampling-time lever orthogonal to the reward.

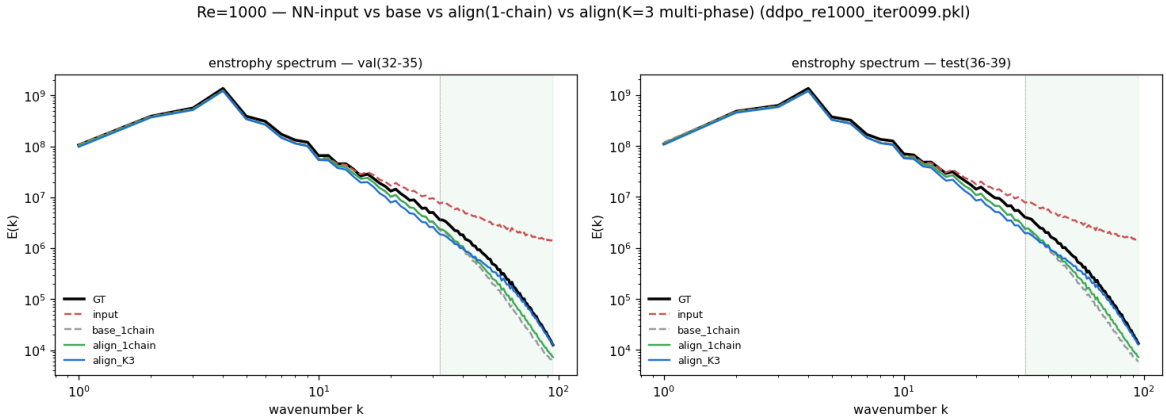


Figure 1. Enstrophy spectra: the NN-fill input carries 7× spurious high-k energy; reconstruction removes it; align+K=3 best approaches GT.

6 Resolution study: where does DDPO pay off? — headline

Clean regular-grid downsampling (every N-th pixel + NN-fill), base model. Placement and reconstruction quality vs observation density.

grid	#pts	%	E-placement	PDE-placement	hik_ret	MSE
16×	256	0.39	0.18	0.11	0.71	0.387
8×	1024	1.56	0.32	0.20	0.46	0.093
random-1024 (task)	1024	1.56	0.34	0.15	0.47	0.172
6×	1849	2.82	0.39	0.25	0.33	0.041
5×	2704	4.13	0.54	0.29	0.33	0.019
4×	4096	6.25	0.81	0.35	0.39	0.019
3×	7396	11.3	0.95	0.47	0.55	0.005

Energy placement crosses 0.6 at ~3,000 points (~4.5× downsampling). The payoff window is ~3k–5k points: there placement ≥ 0.6 , MSE ≈ 0.019 (near-perfect), yet hik_ret dips to ~0.33 — the model is faithful but under-energized, a real deficit the spectral reward can fill with no placement penalty. Below the window the reward adds energy in the wrong places (the 1024-point dead end); above it the input already supplies the high-k. **PDE-residual placement lags** — it never reaches 0.6 even at 11% sampling — so lean on the spectral reward at the payoff resolution and treat PDE as a mild regularizer. At equal budget (1024 pts) a regular grid beats random collocation on MSE (0.093 vs 0.172).

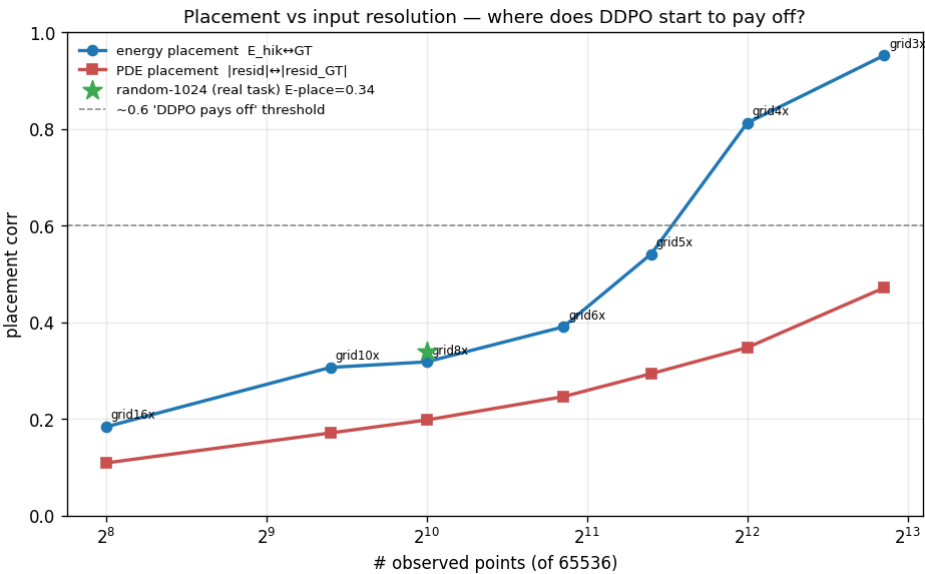


Figure 2. Energy- and PDE-residual placement vs observation count. Energy placement crosses the ~0.6 'DDPO pays off' line near 3,000 points; PDE placement lags throughout.

7 Status and next steps

Running: grid-4× (4096-pt) DDPO finetuning with the align reward — the direct test that DDPO becomes additive (moves MSE and retention together) once placement is solved by the input (base@grid-4×: placement 0.81, hik_ret 0.41, MSE 0.019).

Levers for the placement bottleneck, ranked: (1) observation density — dominant and provable; (2) multi-phase sampling — free at inference; (3) multi-phase finetuning — plausible at 2–3× training cost, test K=2 first; (4) reward engineering — effectively exhausted at 1024 points. A deterministic energy+PDE retrain would regress to the posterior mean and over-smooth like MSE; the generative sampler is what buys sharpness.

Code: `src/ddpo_ft/ (train_claude, eval_ddpo, eval_multiphase, diag_resolution[_sweep], diag_strain, diag_residual_locality, viz_*)`; `base_results/build_extrap_anchor.py`. Detailed narrative in

docs/ddpo_findings.md.